

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Accepts soil and environmental parameters	UI Module	Done	15%
2	Data Validation	Validates user inputs before prediction.	Flask Backend	Done	10%
3	Crop Prediction	Predicts the most suitable crop	ML Prediction Module	Done	25%
4	Model Loading	Loads the trained ML model for crop prediction.	Model Module	Done	10%
5	Dataset Processing	Reads and preprocesses the crop recommendation dataset.	Data Processing Module	Done	10%
6	Result Display	Displays predicted crop to the user	UI Module	Done	15%
7	About Page	Displays project information	Web Module	Done	5%
8	Error Handling	Handles invalid inputs and prediction errors gracefully.	Backend Logic	Done	10%

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Home Page, About Page, Crop Prediction Form, Result Display	HTML5, CSS3, Bootstrap, Flask Templates
2	Backend / Logic	Input Validation, Data Processing, ML Prediction, Flask Backend	Python, Flask, Scikit-learn, Pandas, NumPy
3	Database / Storage	Crop Recommendation Dataset, Trained ML Model	CSV Dataset, Pickle (.pkl), Local Storage
4	API / Integration	Flask Routing, HTTP Request Handling, Template Rendering	Flask API, Jinja2 Templates
5	Security / Authentication	Input Validation, Exception Handling, Error Management	Flask Validation, Python Exception Handling